MyLearna
Plan • Capture • Grow

MATHEMATICS

RATIO AND PROPORTIONAL REASONING

Name: _____

Date: _____

Judge Fairness, Value, and Efficiency Proportionally

Step 10

Elementary (Grades 2–4)

Today I am learning to judge fairness, value, and efficiency by making proportional comparisons.

1. Is It Fair?

Look at each situation. Decide if it is fair.
Tick ✓ Fair or ✗ Not Fair. Explain your thinking.

a) Two friends share 8 cookies.
Sam gets 5 cookies. Alex gets 3 cookies.



Fair: ☐ Not Fair: ☐

Why? _____

b) Three children share 12 stickers equally.
Each child gets 4 stickers.





Fair: ☐ Not Fair: ☐

Why? _____

c) A bag of lollies has 20 lollies.
One child gets 12. Another gets 8.



Fair: ☐ Not Fair: ☐

Why? _____

2. Which Is the Better Value?

Find the unit price (price per item) to decide which is the better value.
Circle the better value.

a) Apples

Option A	Option B
6 apples for $3.60	8 apples for $4.80

b) Milk

Option A	Option B
2 litres for $3.20	3 litres for $4.50

c) Pencils

Option A	Option B
10 pencils for $2.00	15 pencils for $2.85

Remember!
To find unit price:
Unit price = Total cost ÷ Number of items

3. Which Is More Efficient?

Decide which option is more efficient.
Compare the rate (how fast or how much is done per unit).

a) Filling a pool

Option A	Option B
The pool is filled with 300 litres of water in 30 minutes.	The pool is filled with 450 litres of water in 40 minutes.



Which is more efficient? _____ Why? _____

b) Running

Option A	Option B
A runner goes 4 km in 20 minutes.	A runner goes 6 km in 25 minutes.



Which is more efficient? _____ Why? _____

4. Best Deal Overall

Three stores sell the same box of cereal.
Use unit price to find the best deal.

Store	Size of Box	Price	Unit Price (per 100 g)
Store A	375 g	$3.75	$_____
Store B	500 g	$4.00	$_____
Store C	750 g	$5.25	$_____



Which store is the best deal? _____

Why? _____

5. Fair Share in Different Ways

Decide if each distribution is fair.
If not fair, how could it be made fair?

a) 15 marbles are shared between 4 children.
Three children get 4 marbles each.
One child gets 3 marbles.



Fair: ☐ Not Fair: ☐

How could it be made fair? _____

b) 18 slices of pizza are shared between 3 friends.
One friend gets 8 slices. Another gets 6 slices.
The third gets 4 slices.



Fair: ☐ Not Fair: ☐

How could it be made fair? _____

6. Real-Life Choices

Choose the BEST option in each situation.



a) You buy a 1.5 L bottle of juice for $4.20
or a 2 L bottle of juice for $5.00.
Which should you buy? _____ Why? _____

b) You can print photos.

Option A	Option B
20 photos for $6.00	35 photos for $9.10



Which option should you choose? _____ Why? _____

c) A taxi charges $2.50 to start, then $1.60 per kilometre.
Is this fair? Why or why not? _____

7. Challenge Problem

Two groups complete the same job.
Who was more efficient? Show your working.

Group A	Group B
Painted a fence in 6 hours. They painted 54 metres.	Painted the same fence in 8 hours. They painted 72 metres.



Who was more efficient? _____

Show your working.

8. Think and Explain

Explain your thinking.

a) Why is it important to compare unit rates?

b) Give an example from your life where it helps to compare value or efficiency.

Think!
Good decisions come from smart comparisons!



I can...

- ☐ judge fairness using equal shares
- ☐ compare value using unit prices
- ☐ compare efficiency using rates
- ☐ solve problems using proportional reasoning in real life
- ☐ explain my thinking

Key Words

fair	–	equal or without favour
value	–	how good a deal something is
efficient	–	doing the job using the least time or resources
unit price	–	the price for one item or one unit
rate	–	a comparison of two quantities

Great job!

You are using proportional reasoning to make fair, smart and efficient decisions!